Muhammad Haider Zaidi

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github.com/chefhaider Bavaria, Germany 21.06.2000

Technical Skills

Programming: Python, C++, SQL, MATLAB/Simulink, C
Tools/Frameworks: TensorFlow, PyTorch, OpenAI API, Databricks, FastAPI, Docker, Kubernetes, TensorRT, YOLO, AWS SageMaker, Flask, HTML, CSS
Competence: Machine Learning, Statistical Signal Processing, MLOps, Data Engineering, Pattern Recognition, Data Visualization, Automation, Real-time Systems, AI-based System Analysis

Experience

German National Metrology Institute (PTB) — **March 2024 – Present**
Student Assistant Researcher — *Berlin, DE*

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Explored novel architectures and physics-informed hybrid loss functions to enhance model accuracy and stability.
- Collaborated with PhD researchers to publish findings at the FLOMEKO conference.
- Built dashboards for data visualization and model performance monitoring.

Folio3 — **March 2022 – Nov 2023**
Machine Learning Engineer — *California, US (remote)*

- Improved vitiligo segmentation accuracy to 94% and increased processing speed by 5x using advanced AI algorithms; deployed with Docker and FastAPI.
- Developed a real-time tennis foul detection system using TensorRT, reducing compute cost and time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure.
- Conducted educational sessions on machine learning and big data at universities, enhancing knowledge sharing and collaboration.
- Developed POCs leveraging YOLO, few-shot learning, and CNNs for client demonstrations.

Manda GmbH — **March 2024 - June 2024**
Werkstudent Python Developer — *Nürnberg, DE*

- Developed a full-stack website offering AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and large language model (LLM) tools.

Freelance — **July 2020 – March 2022**
Data Engineer — *Remote*

- Developed web scrapers and ETL pipelines for structured and unstructured data extraction and transformation.
- Automated data ingestion and cleaning workflows for client analytics use cases, improving data processing efficiency.

Projects

CAFA 6: Protein Function Prediction **2022**

Deep Learning Competition

- Designed a weighted-probability ensemble integrating DPFunc and DeepGOPlus with AlphaFold for feature extraction and taxonomic bias mitigation.
- Enhanced model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting **2022**

Thesis Project

- Developed a real-time Bitcoin price forecasting system using machine learning models and time-series analysis.
- Implemented data pipelines for real-time data ingestion, processing, and visualization.

Saaz: AI Self-Playing Chitrali Sitar **2023**

Signal Processing, Music Generation

- Collaborated to create the world's first AI-driven self-playing Chitrali Sitar using signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI-driven music technology.

Hugging Face **Open-Source**

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität **Present**

Master of Science in Artificial Intelligence *Erlangen, DE*

- Specialized in Machine Learning, Statistical Signal Processing, and Data-Driven Workflows.
- Conducted research on automated system analysis using AI and pattern recognition techniques.

National University of Computer and Emerging Sciences **Aug 2018 – June 2022**

Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor for academic excellence.
- Focused on machine learning, data engineering, and software development.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera**: TensorFlow Developer Certificate | **Databricks**: Lakehouse Fundamentals | **Datacamp**: Computer Vision in Python